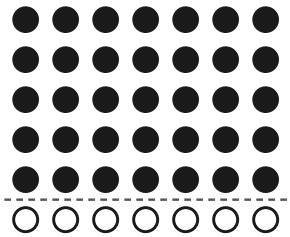
Multiplication Strategies

Break a hard fact into facts you know.

6 × 7: 5 rows and 1 more row



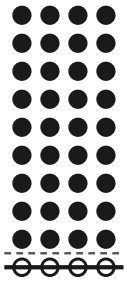
$5 \times 7 = 35$

$1 \times 7 = 7$

$35 + 7 = 42$

×5 is easy: half of ×10.
Then add one more row.

9 × 4: 10 rows, then take one away



$10 \times 4 = 40$

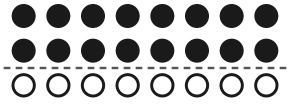
$40 - 4 = 36$

The crossed-out row
is the group taken away.

A strategy for every fact

- **×3** = ×2 and one more group
- **×4** = double, then double again
- **×6** = ×5 and one more group
- **×7** = ×5 and ×2
- **×8** = ×4, then double
- **×9** = ×10, take one group away

Your turn!



$2 \times 8 =$ 16

$1 \times 8 =$ 8

16 + 8 = 24

For teachers

Goal: use ×2, ×5 and ×10 to find the other facts by breaking an array apart (3.OA.5, 3.OA.7). Draw the dashed line, solve each part, then add (or take away for ×9). Practice a few facts a day until they are quick.

Multiplication Strategies: Practice

Break the array. Solve each part. Put them together.

Part 1 Times 3 ×3 = ×2 and one more group.

1) $3 \times 6 = 18$



$2 \times 6 = 12$

$1 \times 6 = 6$

$12 + 6 = 18$

2) $3 \times 4 = 12$



$2 \times 4 = 8$

$1 \times 4 = 4$

$8 + 4 = 12$

3) $3 \times 9 = 27$



$2 \times 9 = 18$

$1 \times 9 = 9$

$18 + 9 = 27$

4) $3 \times 7 = 21$



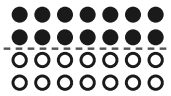
$2 \times 7 = 14$

$1 \times 7 = 7$

$14 + 7 = 21$

Part 2 Times 4 ×4 = double, then double again.

5) $4 \times 7 = 28$



$2 \times 7 = 14$

$2 \times 7 = 14$

$14 + 14 = 28$

6) $4 \times 3 = 12$



$2 \times 3 = 6$

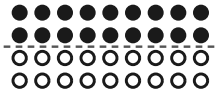
$2 \times 3 = 6$

$6 + 6 = 12$

Multiplication Strategies: Practice (continued)

Part 2 Times 4 (continued)

7) $4 \times 9 =$ **36**

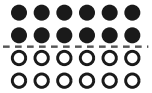


$2 \times 9 =$ **18**

$2 \times 9 =$ **18**

18 + **18** = **36**

8) $4 \times 6 =$ **24**



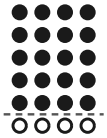
$2 \times 6 =$ **12**

$2 \times 6 =$ **12**

12 + **12** = **24**

Part 3 Times 6 ×6 = ×5 and one more group.

9) $6 \times 4 =$ **24**

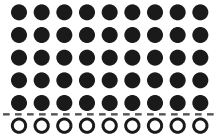


$5 \times 4 =$ **20**

$1 \times 4 =$ **4**

20 + **4** = **24**

10) $6 \times 9 =$ **54**

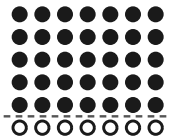


$5 \times 9 =$ **45**

$1 \times 9 =$ **9**

45 + **9** = **54**

11) $6 \times 7 =$ **42**



$5 \times 7 =$ **35**

$1 \times 7 =$ **7**

35 + **7** = **42**

12) $6 \times 3 =$ **18**



$5 \times 3 =$ **15**

$1 \times 3 =$ **3**

15 + **3** = **18**

Multiplication Strategies: Practice (continued)

Part 4 Times 9 ×9 = ×10, take one group away.

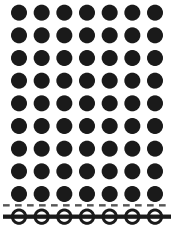
13) $9 \times 3 = \boxed{27}$



$10 \times 3 = \boxed{30}$

$\boxed{30} - 3 = \boxed{27}$

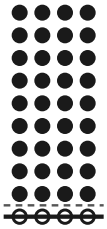
14) $9 \times 7 = \boxed{63}$



$10 \times 7 = \boxed{70}$

$\boxed{70} - 7 = \boxed{63}$

15) $9 \times 4 = \boxed{36}$



$10 \times 4 = \boxed{40}$

$\boxed{40} - 4 = \boxed{36}$

Part 5 Times 8 ×8 = ×4, then double.

16) $8 \times 3 = \boxed{24}$

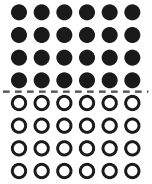


$4 \times 3 = \boxed{12}$

$4 \times 3 = \boxed{12}$

$\boxed{12} + \boxed{12} = \boxed{24}$

17) $8 \times 6 = \boxed{48}$



$4 \times 6 = \boxed{24}$

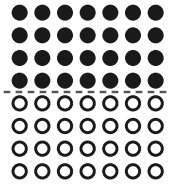
$4 \times 6 = \boxed{24}$

$\boxed{24} + \boxed{24} = \boxed{48}$

Multiplication Strategies: Practice (continued)

Part 5 Times 8 (continued)

18) $8 \times 7 = 56$



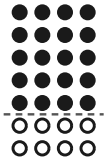
$4 \times 7 = 28$

$4 \times 7 = 28$

$28 + 28 = 56$

Part 6 Times 7 ×7 = ×5 and ×2.

19) $7 \times 4 = 28$

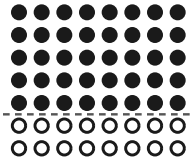


$5 \times 4 = 20$

$2 \times 4 = 8$

$20 + 8 = 28$

20) $7 \times 8 = 56$

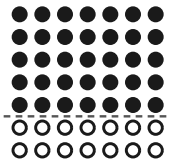


$5 \times 8 = 40$

$2 \times 8 = 16$

$40 + 16 = 56$

21) $7 \times 7 = 49$



$5 \times 7 = 35$

$2 \times 7 = 14$

$35 + 14 = 49$

Part 7 Times 3 No help this time.

22) $3 \times 7 = 21$

23) $7 \times 3 = 21$

24) $3 \times 9 = 27$

25) $9 \times 3 = 27$

26) $3 \times 6 = 18$

27) $4 \times 3 = 12$

Multiplication Strategies: Practice (continued)

Part 7 Times 3 (continued)

28) $3 \times 8 = 24$

29) $8 \times 3 = 24$

30) $3 \times 4 = 12$

31) $6 \times 3 = 18$

32) $3 \times 3 = 9$

Part 8 Times 4 No help this time.

33) $4 \times 7 = 28$

34) $6 \times 4 = 24$

35) $4 \times 9 = 36$

36) $7 \times 4 = 28$

37) $8 \times 4 = 32$

38) $4 \times 8 = 32$

Multiplication Strategies: Practice (continued)

Part 8 Times 4 (continued)

39) $9 \times 4 = 36$

40) $4 \times 4 = 16$

41) $4 \times 6 = 24$

Part 9 Times 6 No help this time.

42) $9 \times 6 = 54$

43) $6 \times 8 = 48$

44) $7 \times 6 = 42$

45) $6 \times 6 = 36$

46) $8 \times 6 = 48$

47) $6 \times 7 = 42$

48) $6 \times 9 = 54$

Multiplication Strategies: Practice (continued)

Part 10 Times 7, 8 and 9 No help this time.

49)	$8 \times 7 = 56$	50)	$7 \times 7 = 49$
51)	$9 \times 8 = 72$	52)	$8 \times 9 = 72$
53)	$9 \times 9 = 81$	54)	$7 \times 8 = 56$
55)	$9 \times 7 = 63$	56)	$7 \times 9 = 63$
57)	$8 \times 8 = 64$		

Part 11 Break it apart Use a fact you know, then add the rest.

58) $3 \times 9 = 3 \times 5 + 3 \times 4 = 27$

59) $7 \times 8 = 7 \times 5 + 7 \times 3 = 56$

60) $4 \times 9 = 4 \times 5 + 4 \times 4 = 36$

Multiplication Strategies: Practice (continued)

Part 11 Break it apart (continued)

61) $9 \times 6 = 9 \times 5 + 9 \times 1 =$ **54**

62) $8 \times 8 = 8 \times 5 + 8 \times 3 =$ **64**

63) $6 \times 8 = 6 \times 5 + 6 \times 3 =$ **48**

64) $4 \times 7 = 4 \times 5 + 4 \times 2 =$ **28**

65) $3 \times 6 = 3 \times 5 + 3 \times 1 =$ **18**

66) $8 \times 6 = 8 \times 5 + 8 \times 1 =$ **48**

67) $9 \times 9 = 9 \times 5 + 9 \times 4 =$ **81**

68) $8 \times 9 = 8 \times 5 + 8 \times 4 =$ **72**

69) $3 \times 8 = 3 \times 5 + 3 \times 3 =$ **24**

70) $6 \times 6 = 6 \times 5 + 6 \times 1 =$ **36**

71) $7 \times 7 = 7 \times 5 + 7 \times 2 =$ **49**

72) $6 \times 7 = 6 \times 5 + 6 \times 2 =$ **42**

Multiplication Strategies: Practice (continued)

Part 11 Break it apart (continued)

73) $4 \times 6 = 4 \times 5 + 4 \times 1 =$ **24**

74) $8 \times 7 = 8 \times 5 + 8 \times 2 =$ **56**

75) $7 \times 9 = 7 \times 5 + 7 \times 4 =$ **63**

76) $4 \times 8 = 4 \times 5 + 4 \times 3 =$ **32**

77) $9 \times 7 = 9 \times 5 + 9 \times 2 =$ **63**

78) $6 \times 9 = 6 \times 5 + 6 \times 4 =$ **54**

79) $9 \times 8 = 9 \times 5 + 9 \times 3 =$ **72**

80) $7 \times 6 = 7 \times 5 + 7 \times 1 =$ **42**

81) $3 \times 7 = 3 \times 5 + 3 \times 2 =$ **21**

Multiplication Strategies: Practice (continued)

Part 12 Remember? Problems from earlier packets.

82) 6 groups of 5 = **30**

83) half past 2 = **2:30**

84) 3 dimes + 4 nickels = **50¢**

85) 8 groups of 2 = **16**

86) 502 − 267 = **235**

87) 45 cm + 38 cm = **83** cm

Part 13 Word problems Find the groups and how many in each.

88) A garden has 6 rows with 7 carrots in each row. How many carrots are there?

Number sentence **6 × 7 = 42**

Answer **42** carrots

Draw or show your work

89) A spider has 8 legs. How many legs do 4 spiders have?

Number sentence **4 × 8 = 32**

Answer **32** legs

Draw or show your work

90) A pack has 9 markers. How many markers are in 3 packs?

Number sentence **3 × 9 = 27**

Answer **27** markers

Draw or show your work

Multiplication Strategies: Practice (continued)

Part 13 Word problems (continued)

91) There are 7 days in a week. How many days are in 8 weeks?

Number sentence **8 × 7 = 56**

Answer **56** days

Draw or show your work

92) Challenge: A theater has 9 rows of 8 seats. 15 seats are empty. How many seats are full?

Number sentence **9 × 8 = 72, 72 − 15 = 57**

Answer **57** seats

Draw or show your work

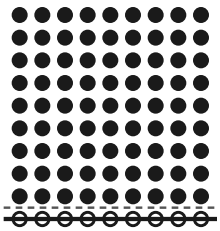
Check-In: Multiplication Strategies

Show what you know. No help this time.

Mastery: 9 of 11 (80%)

Part 1 Break it apart

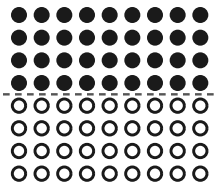
1) $9 \times 9 = 81$



$10 \times 9 = 90$

$90 - 9 = 81$

2) $8 \times 9 = 72$



$4 \times 9 = 36$

$4 \times 9 = 36$

$36 + 36 = 72$

Part 2 Multiply

3) $3 \times 6 = 18$

4) $9 \times 9 = 81$

5) $7 \times 3 = 21$

6) $4 \times 9 = 36$

7) $7 \times 6 = 42$

8) $6 \times 6 = 36$

9) $3 \times 3 = 9$

10) $6 \times 9 = 54$

11) $7 \times 7 = 49$